

Final Report

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1. Abstract

2. Introduction

Total fertility rates (TFRs) are key population indicators that are widely monitored because they influence age structure and population growth (Wilmoth et al.2022; World Health Organisation). Globally, fertility rates have declined over recent decades (Chen et al. 2018; Shair et al. 2024; Vanella 2024). In Singapore, the total fertility rate has dropped below the replacement level required for a population to replace itself across generations (Jones & Hamid 2015; Tan, 2024; Wilmoth et al. 2022). Therefore, forecasting TFR and total live births (TLB) may provide insight into future fertility patterns and population change in Singapore. Preliminary exploratory data analysis indicated a long-term decline in TFR and TLB with periodic fluctuations, motivating the use of forecasting models capable of capturing these patterns.

Data from 1960 to 2024 on TLB and TFR were obtained from the Singapore Department of Statistics (2023). The research question addressed in this report is: Can ARIMA models accurately forecast Singapore’s TLB and TFR while capturing short-term fluctuations?

This report is structured as follows. Part 3 describes the fertility decline in Singapore, Part 4 describes the process and workflow of obtaining viable ARIMA models,

3. Literature Review

Around 1965, Singapore introduced policies aimed at reducing population growth, which contributed to subsequent declines in fertility rates (Chen et al. 2018; Goodkind 1996; Singh et al. 1988; Vanella 2024). Despite this long-term decline, there may be cyclical patterns in TFR and TLB. These fluctuations may be associated with cultural preferences surrounding Lunar zodiac Dragon years (1964, 1976, 1988, 2000, 2012 and 2024), which are considered especially auspicious years and have been associated with higher fertility rates and live births (Goodkind 1996; Jones & Hamid 2015). Forecasting such series therefore requires models that can capture both long-term trends and short-term variation. Among the various models for time series forecasting, ARIMA models are particularly flexible because they allow selection from a broad class of models rather than imposing a single model form on every time series (Kolambe & Arora 2024; Newbold 1983). Consequently, ARIMA models were selected as the forecasting approach for Singapore’s TFR and TLB.

4. Methods

For both TFR and TLB, an identical modelling workflow was applied. The analysis focused on the period from 1960-2012 for model fitting, with observations 2013-2024 reserved for evaluating forecast accuracy. Initial plots were used to examine the presence of trends, seasonality, and other notable patterns, which indicated a long-term decreasing trend in both series. Several transformations were considered including

logarithmic transformation and first- and second-order differencing. Stationarity was assessed using the KPSS test. For each transformed series, autocorrelation (ACF) and partial autocorrelation (PACF) plots were used to find candidate ARIMA models. The candidate ARIMA models were then fitted and the residual ACF and PACF were examined to assess whether the residuals resembled white noise. Forecast accuracy was evaluated using mean square error (MSE) and mean absolute error (MAE) while model selection was guided by Akaike's Information Criterion (AIC) and the corrected Akaike's Information Criterion (AICc). This procedure was applied consistently to both series to ensure comparability of results.

5. Results

Table 1. TFR - no log transformation

Model	AIC	AICc	MSE	MAE
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547

Table 2. TFR - log transformed

Model	AIC	AICc	MSE	MAE
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933

Table 3. TLB - no log transformation

Model	AIC	AICc	MSE	MAE
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802

Table 4. TLB - log transformed

Log Transformed TLB Model	AIC	AICc	MSE	MAE
ARIMA(15,1,0)	-145.56	-130.02	2239721240	47245.34
ARIMA(12,1,3)	-139.69	-124.15	1752591080	41851.92

6. Discussion

For TFR, ARIMA(16,1,0) has the lowest AIC (-63.99) as the initial ARIMA model but higher MSE and MAE compared to the other models. However, the KPSS test shown that the data needed to be differenced twice. ARIMA(12,2,3) has the lowest AICc (-46.57) and relatively lower MSE and MAE which shows that the data were best to be differenced.

For the log transformed data, ARIMA(13,1,3) has the lowest AIC (-136) but the MSE and MAE were not as low as ARIMA(13,2,2) which has the lowest AICc (-116.39) and lowest MSE (0.007426) and lowest MAE (0.06827) among all TFR, non-log transformed and log transformed models overall. This indicated the data required to be log transformed and differenced.

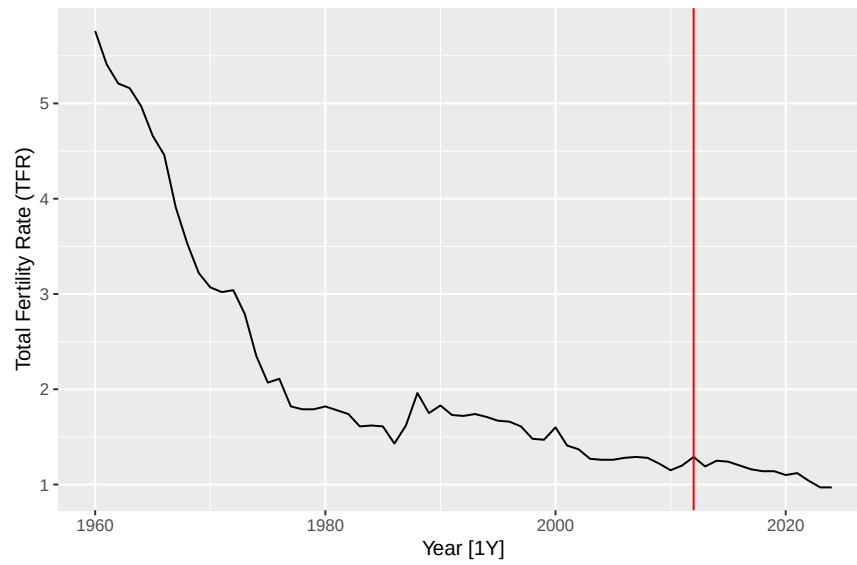
Conclusion

References

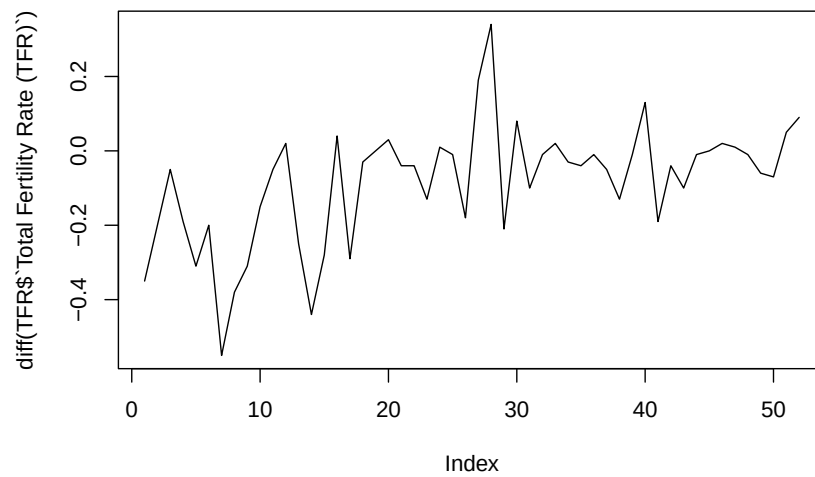
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- World Health Organization. *Total fertility rate (per woman)*. Accessed June 7, 2026, from <https://www.who.int/data/gho/indicator-metadata-registry/imr-details/123>

Appendices

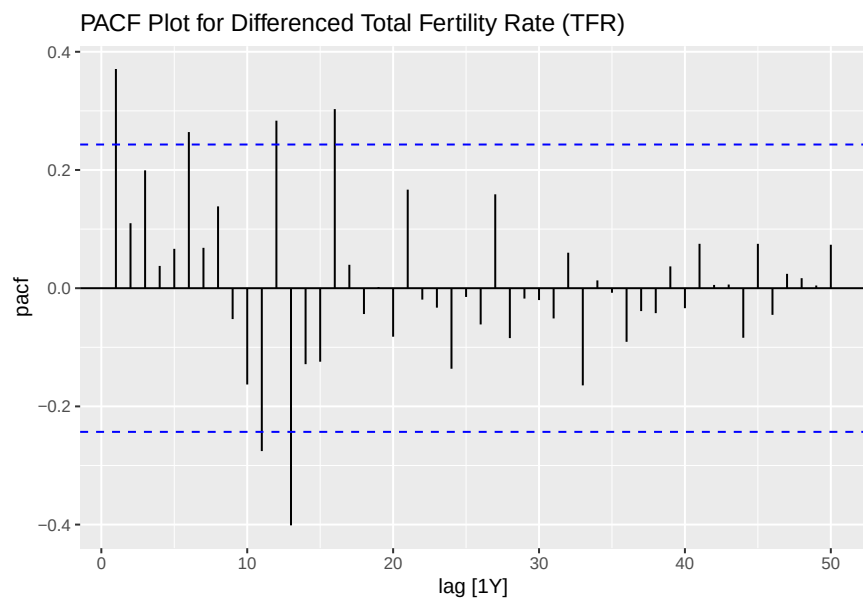
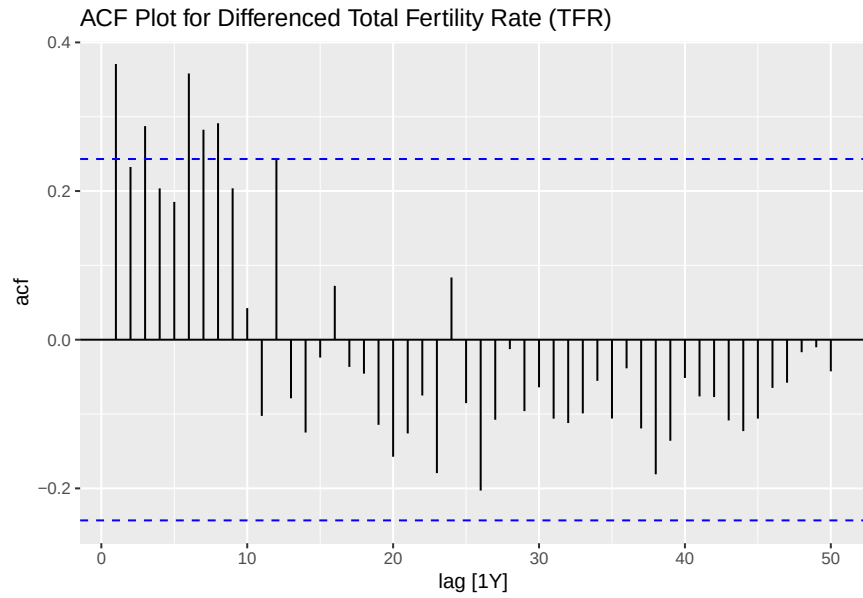
Figure 1. Initial Plot of Total Fertility Rate (TFR)



Plot of Differenced Total Fertility Rate (TFR) until 2012

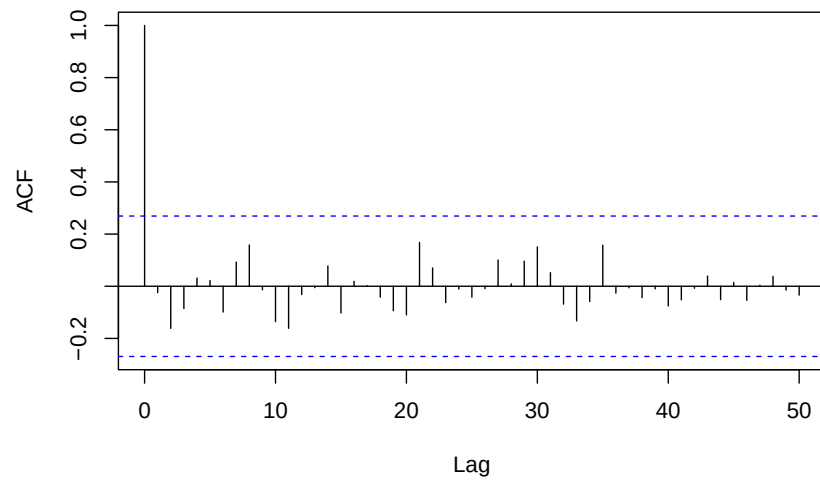


```
##      kpss_stat kpss_pvalue
##      0.8412374  0.0100000
```

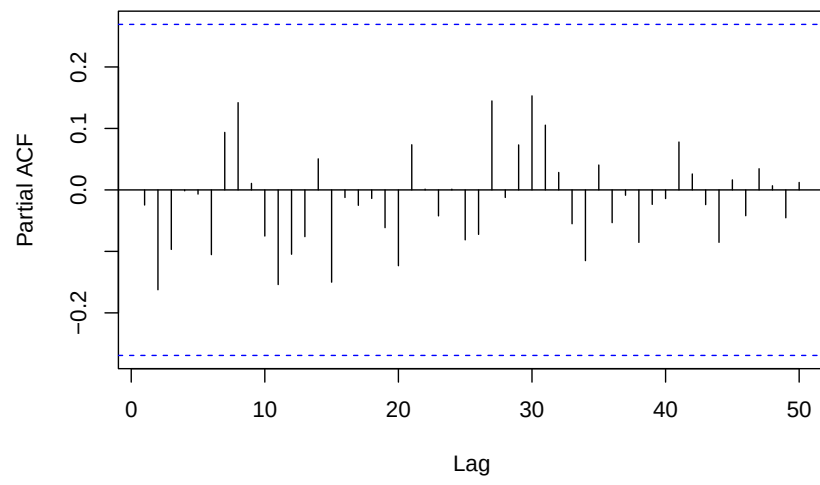


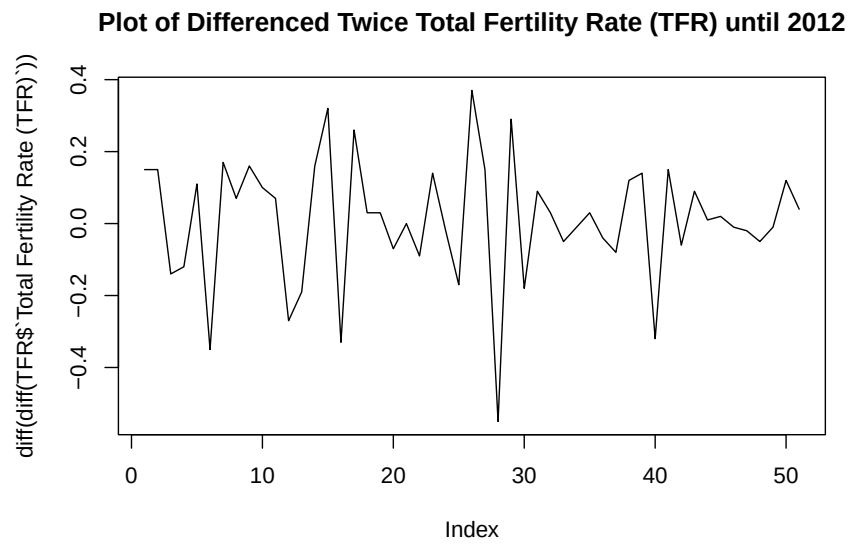
```
## Series: TFR_ts
## ARIMA(16,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.3647 -0.1061  0.4577 -0.2586  0.2984  0.2452  0.2006  0.1578 -0.0293
## s.e.  0.1240  0.1289  0.1246  0.1128  0.0935  0.0832  0.0952  0.0928  0.0996
##      ar10     ar11     ar12     ar13     ar14     ar15     ar16
##      -0.0306 -0.3594  0.3889 -0.6048 -0.0419 -0.3361  0.5135
## s.e.  0.0973  0.0913  0.1039  0.1066  0.1350  0.1295  0.1397
##
## sigma^2 = 0.009716: log likelihood = 48.99
## AIC=-63.99  AICc=-45.99  BIC=-30.82
```

ACF of Residuals for ARIMA(16,1,0) for TFR

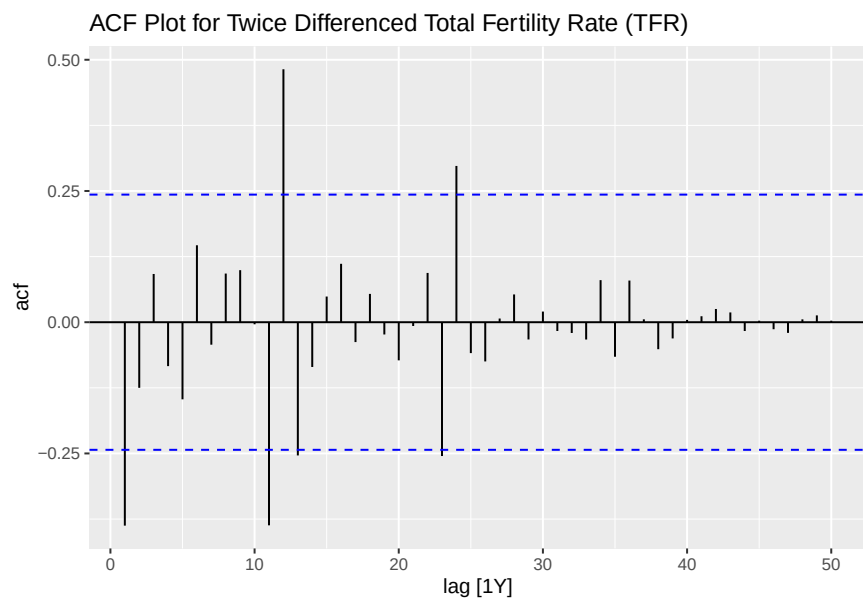


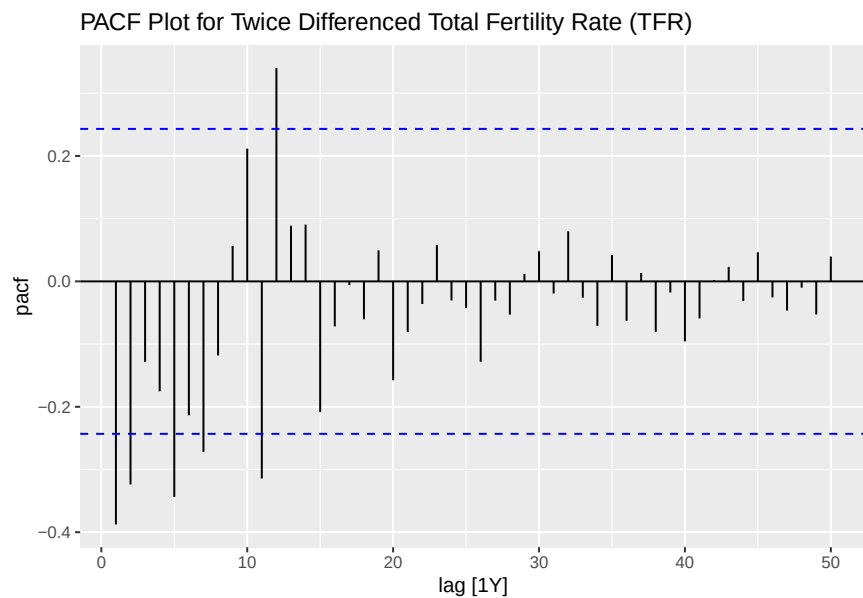
PACF of Residuals for ARIMA(16,1,0) for TFR





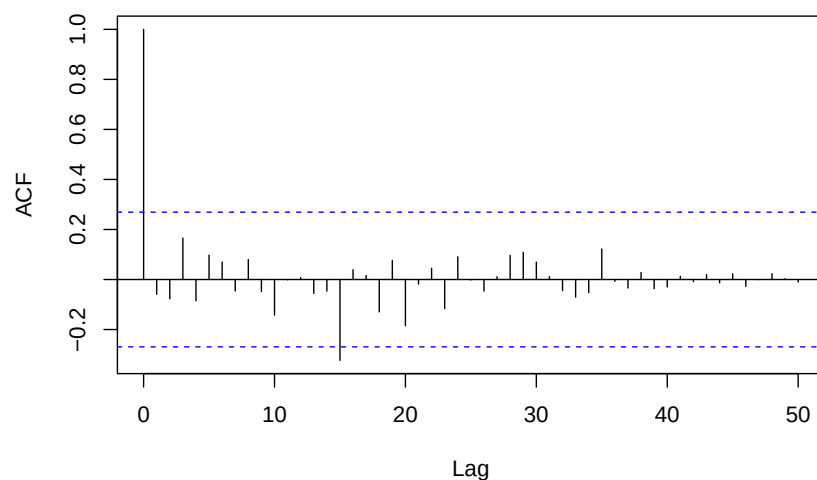
```
##      kpss_stat kpss_pvalue
## 0.04262877 0.10000000
```



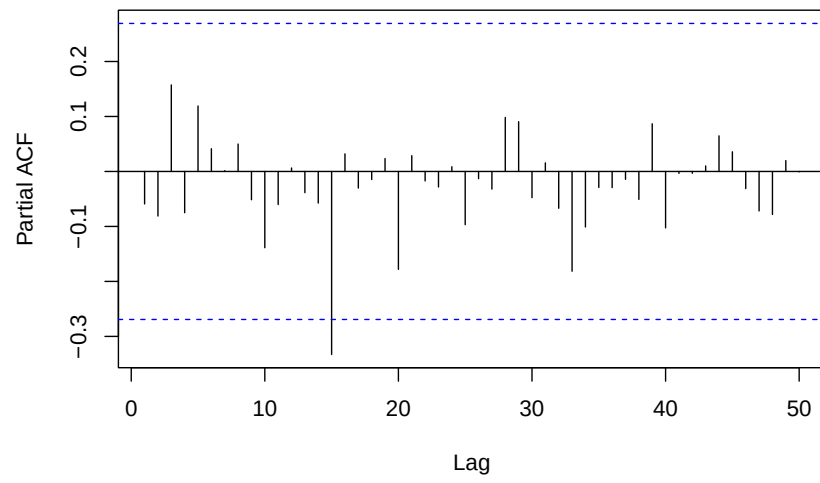


```
## Series: TFR_ts
## ARIMA(12,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      -0.6234 -0.7441 -0.5813 -0.5700 -0.4163 -0.1741  0.0462  0.3482
## s.e.   0.1203  0.1494  0.1673  0.1676  0.1705  0.1772  0.1846  0.1847
##      ar9      ar10     ar11     ar12
##      0.4206  0.4880  0.1609  0.5040
## s.e.   0.1845  0.1783  0.1540  0.1205
##
## sigma^2 = 0.01286: log likelihood = 41.78
## AIC=-57.55  AICc=-47.72  BIC=-32.44
```

ACF of Residuals for ARIMA(12,2,0) for TFR



PACF of Residuals for ARIMA(12,2,0) for TFR

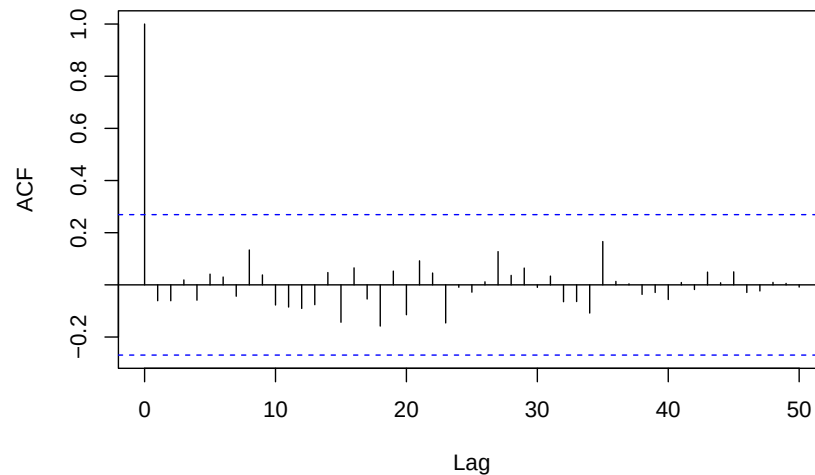


```

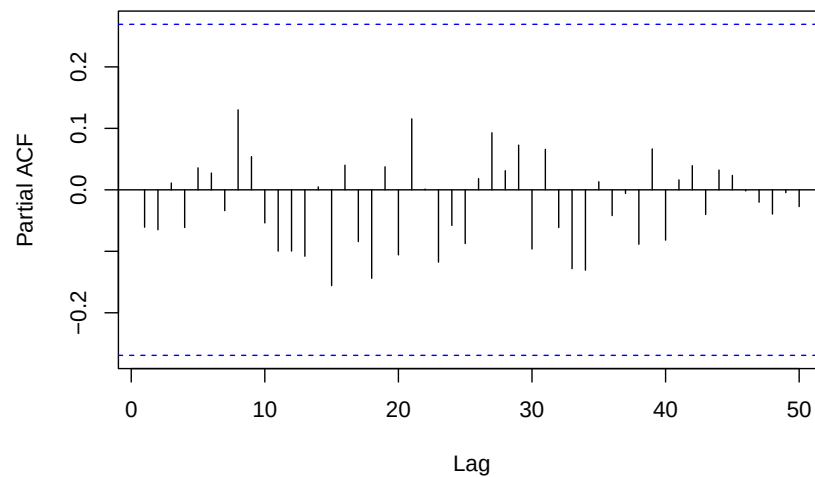
## Series: TFR_ts
## ARIMA(12,2,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.4622 -0.7606 -0.9273 -0.8808 -0.6967 -0.4124 -0.2312  0.0490
## s.e.   0.1091  0.1266  0.1643  0.2032  0.2376  0.2529  0.2548  0.2404
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1578  0.3268  0.1426  0.6463 -0.1402  0.1411  0.8593
## s.e.   0.2067  0.1645  0.1248  0.1005  0.1237  0.1262  0.1309
##
## sigma^2 = 0.009277: log likelihood = 47.29
## AIC=-62.57  AICc=-46.57  BIC=-31.66

```

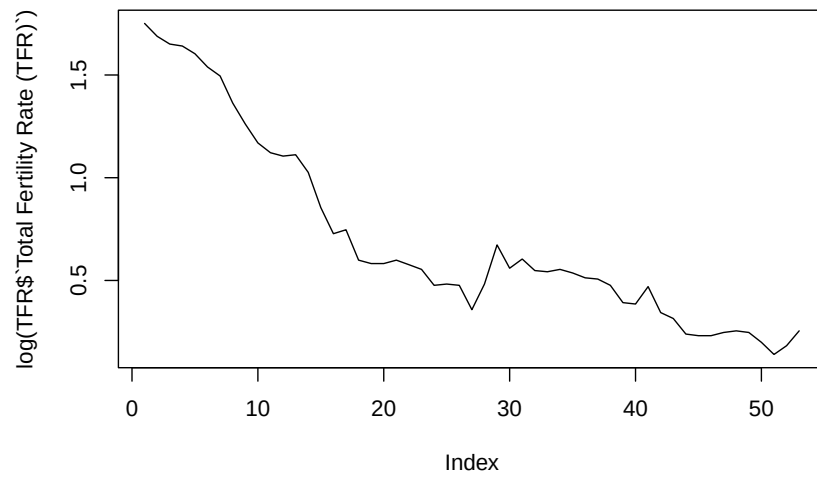
ACF of Residuals for ARIMA(12,2,3) for TFR



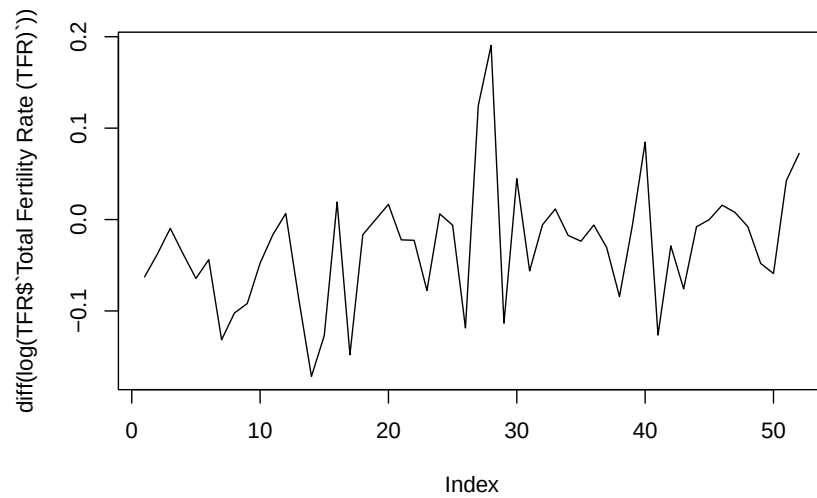
PACF of Residuals for ARIMA(12,2,3) for TFR



Plot of Log Transformation for Total Fertility Rate (TFR) until 2012

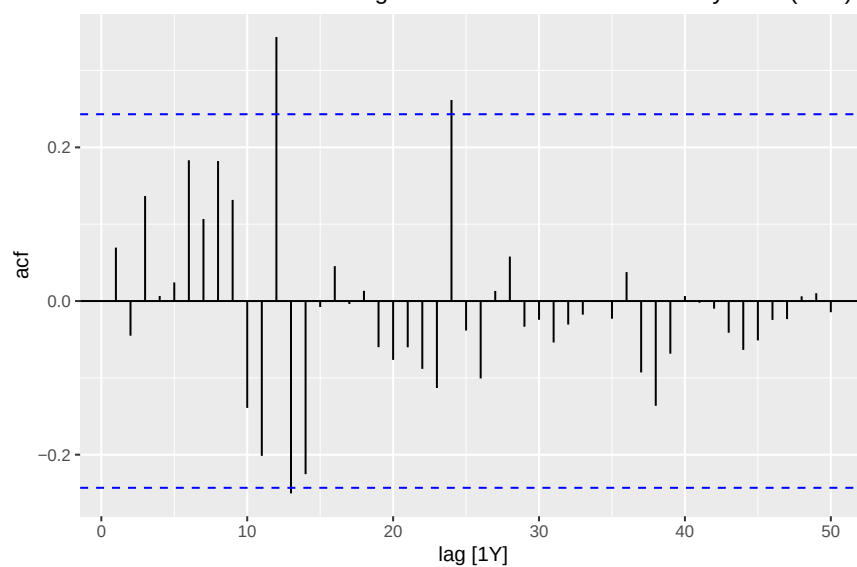


Plot of Differenced Log Transformation for Total Fertility Rate (TFR) until 2012

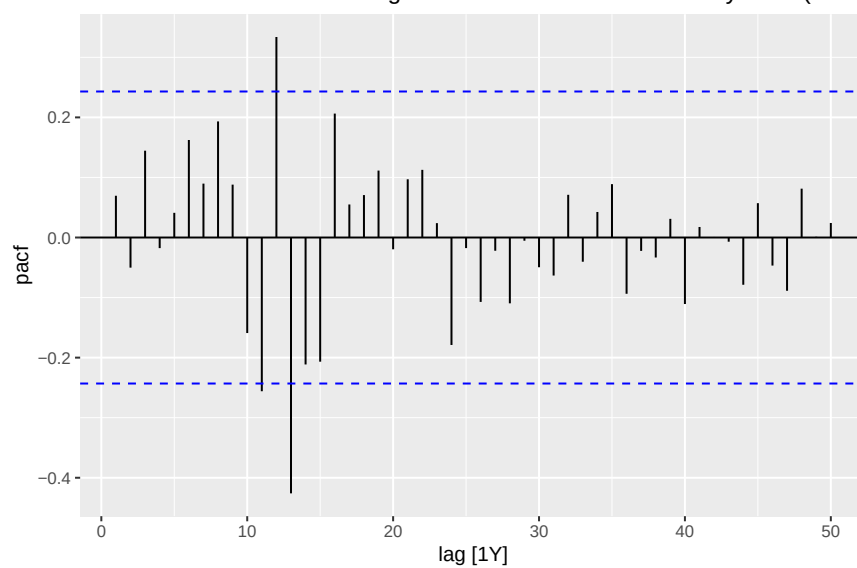


```
##      kpss_stat kpss_pvalue
## 0.50484854 0.04057465
```

ACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)

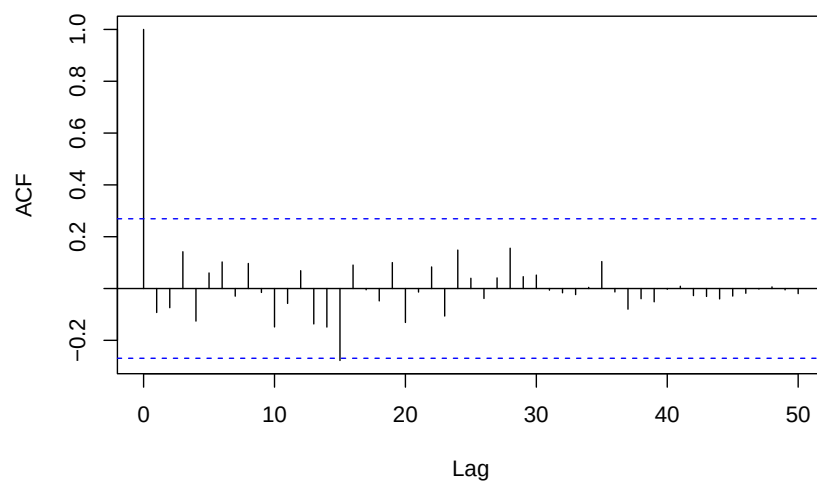


PACF Plot for Differenced Log Transformation of Total Fertility Rate (TFR)

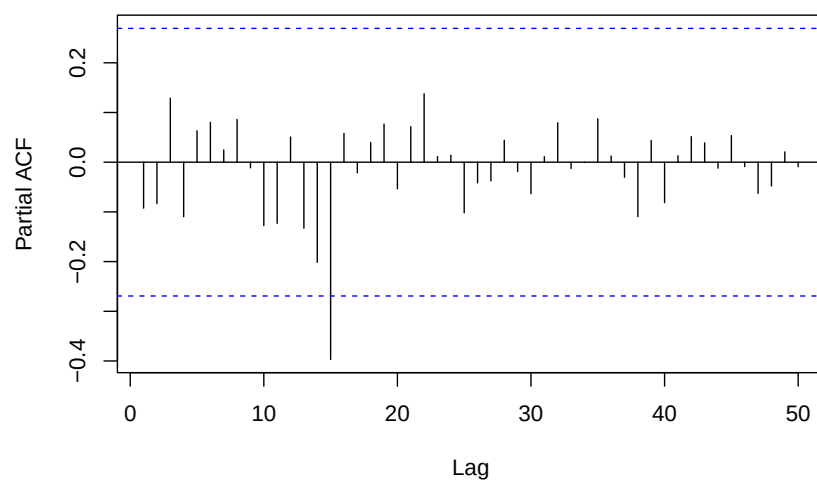


```
## Series: log(TFR_ts)
## ARIMA(13,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2629 -0.0799  0.1194  0.0024  0.1349  0.1715  0.1679  0.2392  0.0298
## s.e.  0.1215  0.1145  0.1065  0.1064  0.1079  0.0974  0.1023  0.1036  0.1106
##      ar10     ar11     ar12     ar13
##      -0.0307 -0.2349  0.3587 -0.4375
## s.e.   0.1130  0.1084  0.1170  0.1258
##
## sigma^2 = 0.00323: log likelihood = 79.9
## AIC=-131.8  AICc=-120.45  BIC=-104.48
```

ACF of Residuals for ARIMA(13,1,0) for TFR with log transformation



PACF of Residuals for ARIMA(13,1,0) for TFR with log transformation

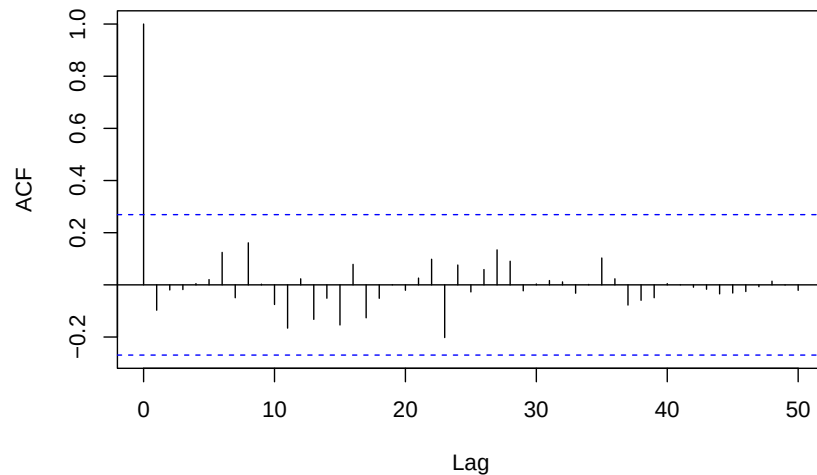


```

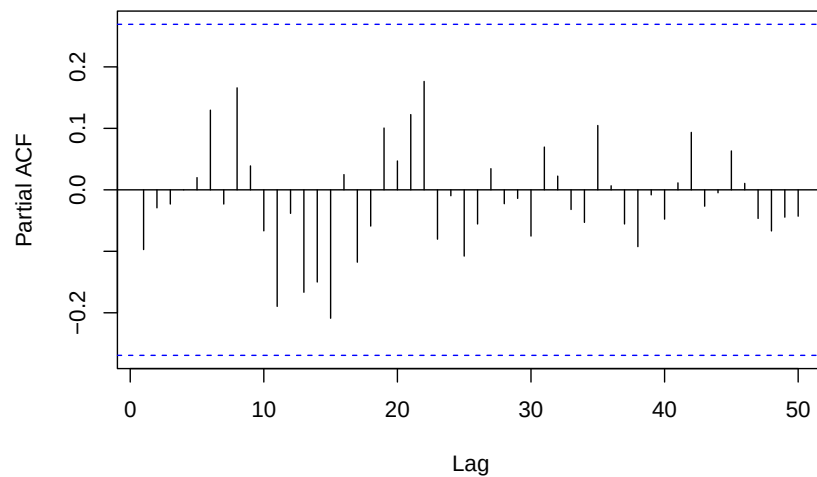
## Series: log(TFR_ts)
## ARIMA(13,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.2783 -0.2434 -0.2961 -0.0013  0.1400  0.2218  0.1567  0.2497  0.0839
## s.e.  0.1256  0.1019  0.1096  0.1094  0.1091  0.0975  0.1024  0.1087  0.1147
##      ar10     ar11     ar12     ar13     ma1      ma2      ma3
##      0.0722 -0.0719  0.4674 -0.5235  0.0903  0.1376  0.9749
## s.e.  0.1170  0.1097  0.1005  0.1256  0.0271  0.1085  0.1089
##
## sigma^2 = 0.00232:  log likelihood = 85
## AIC=-136   AICc=-118   BIC=-102.83

```

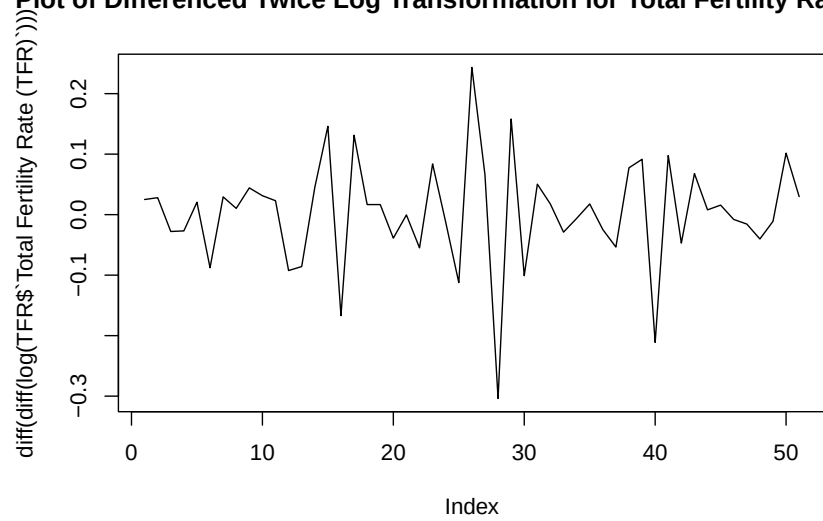
ACF of Residuals for ARIMA(13,1,3) for TFR with log transformation



PACF of Residuals for ARIMA(13,1,3) for TFR with log transformation

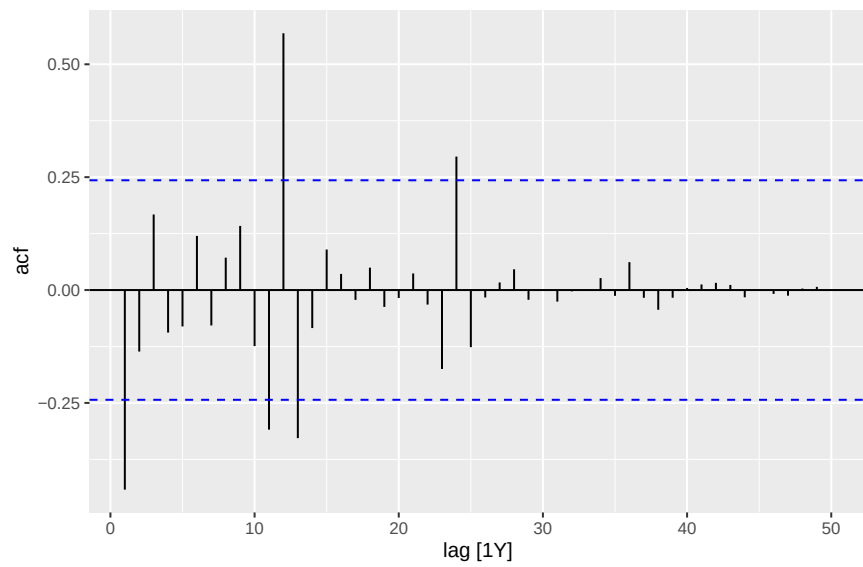


Plot of Differenced Twice Log Transformation for Total Fertility Rate (T

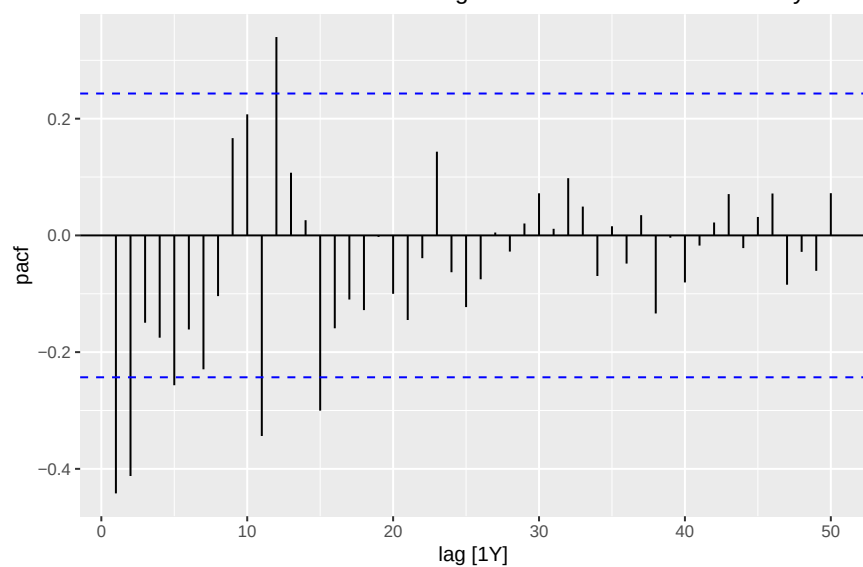


```
##      kpss_stat kpss_pvalue
## 0.04760604 0.10000000
```

ACF Plot for Twice Differenced Log Transformation of Total Fertility Rate (

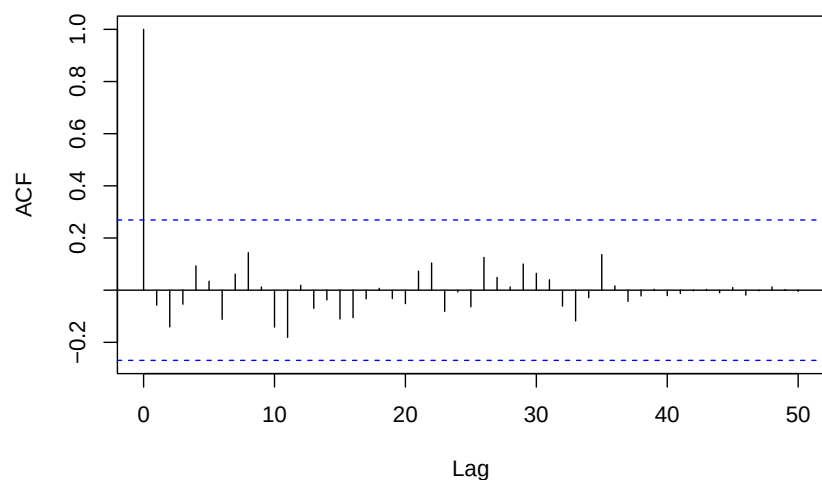


PACF Plot for Twice Differenced Log Transformation of Total Fertility Rate (

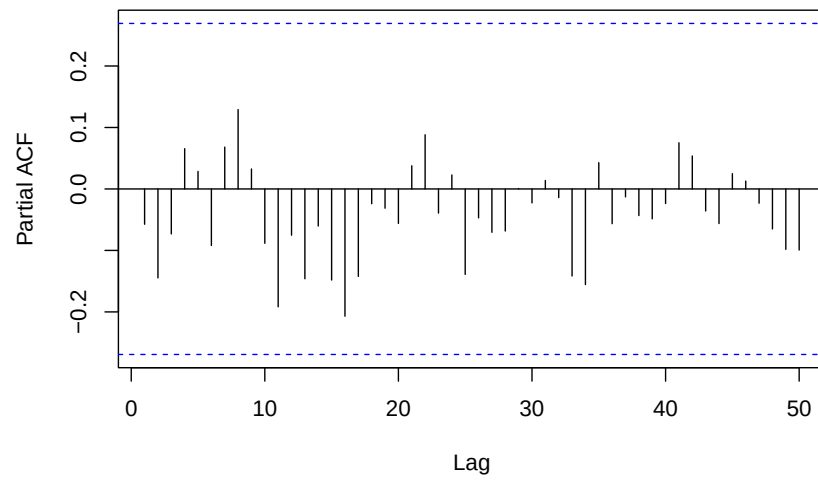


```
## Series: log(TFR_ts)
## ARIMA(15,2,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      -0.7137 -0.7004 -0.3372 -0.5826 -0.3331 -0.1103  0.0628  0.2087
## s.e.   0.1218  0.1571  0.1913  0.1824  0.2031  0.2111  0.2146  0.2149
##      ar9      ar10     ar11     ar12     ar13     ar14     ar15
##      0.1864  0.1142 -0.1162  0.3260 -0.1661 -0.2596 -0.4876
## s.e.   0.2152  0.2137  0.2042  0.1855  0.1901  0.1618  0.1289
##
## sigma^2 = 0.002602: log likelihood = 82.83
## AIC=-133.67  AICc=-117.67  BIC=-102.76
```

ACF of Residuals for ARIMA(15,2,0) for TFR with log transformation

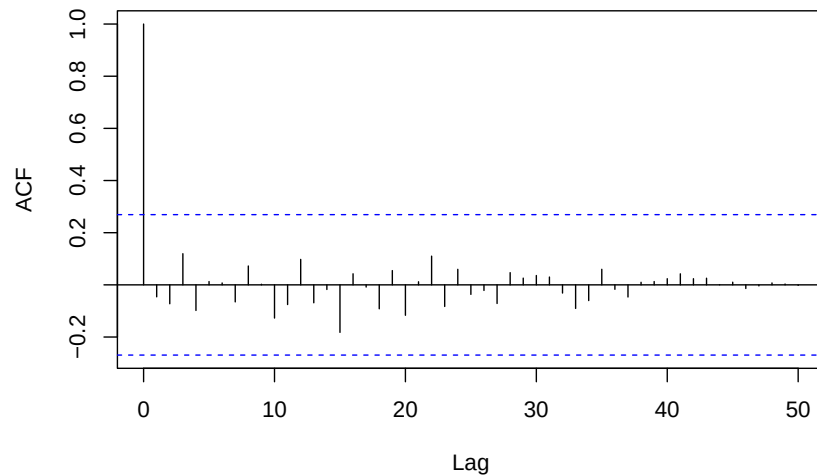


PACF of Residuals for ARIMA(15,2,0) for TFR with log transformatio

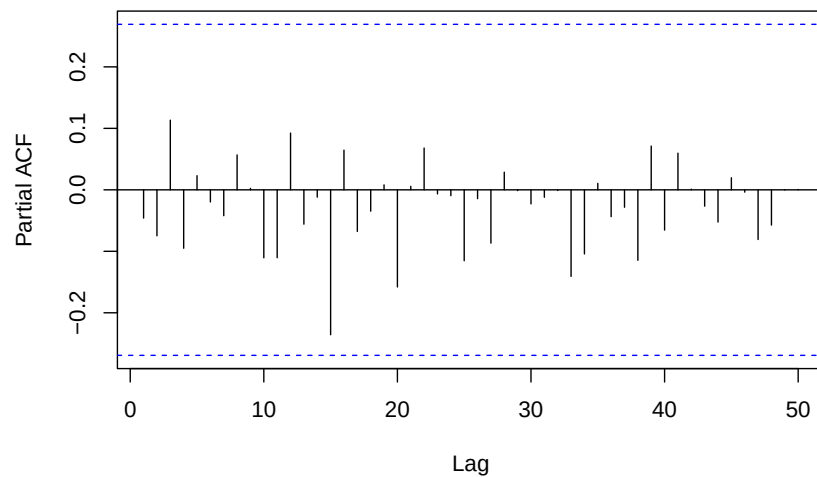


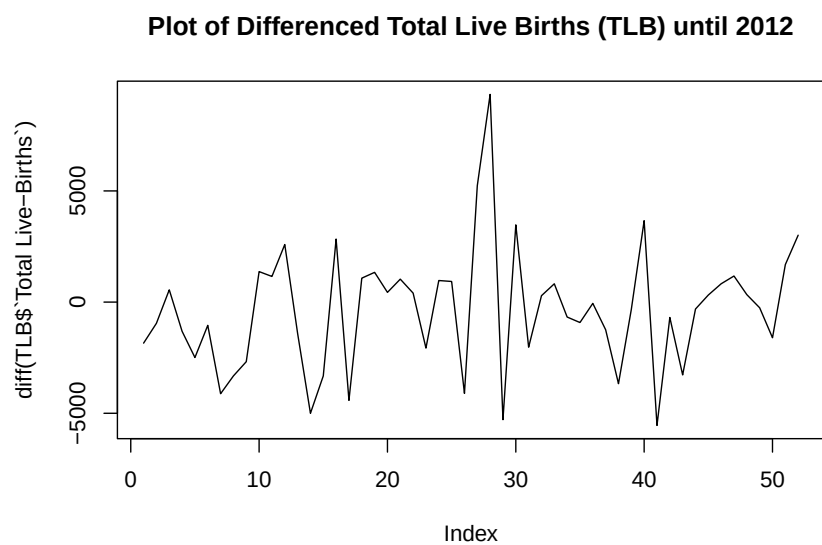
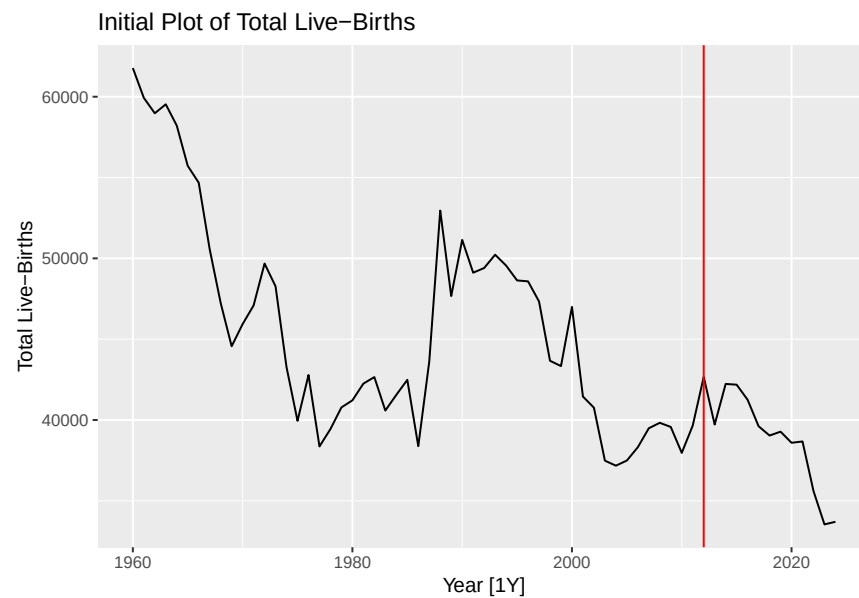
```
## Series: log(TFR_ts)
## ARIMA(13,2,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6514 -0.2412 -0.0063 -0.1608  0.0605  0.0715  0.0280  0.1159
## s.e.  0.1194  0.1202  0.1193  0.1180  0.1158  0.1095  0.1099  0.1125
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.1105 -0.0649 -0.1966  0.4240 -0.6274 -1.6346  0.7851
## s.e.  0.1156  0.1159  0.1124  0.1126  0.0987  0.1446  0.1392
##
## sigma^2 = 0.002717: log likelihood = 82.19
## AIC=-132.39  AICc=-116.39  BIC=-101.48
```

ACF of Residuals for ARIMA(13,2,2) for TFR with log transformation

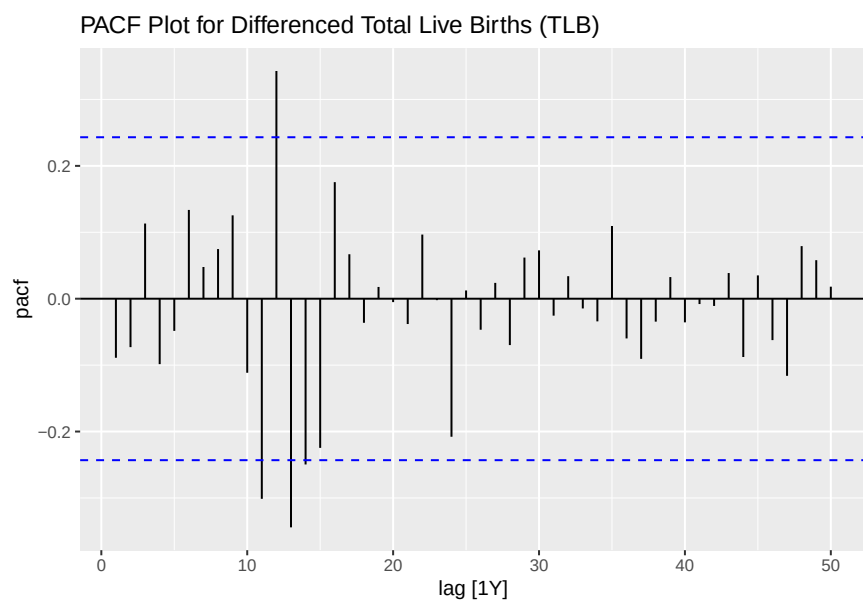
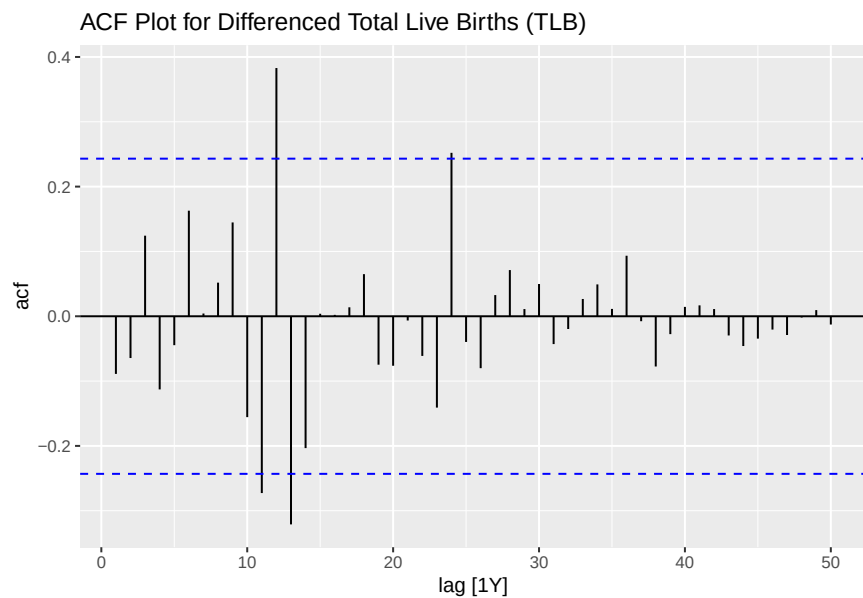


PACF of Residuals for ARIMA(13,2,2) for TFR with log transformation



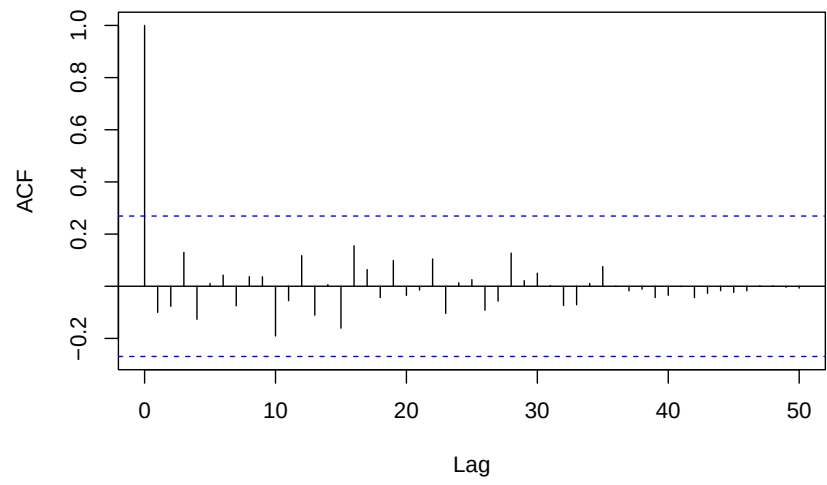


```
##      kpss_stat kpss_pvalue
## 0.2126452    0.1000000
```

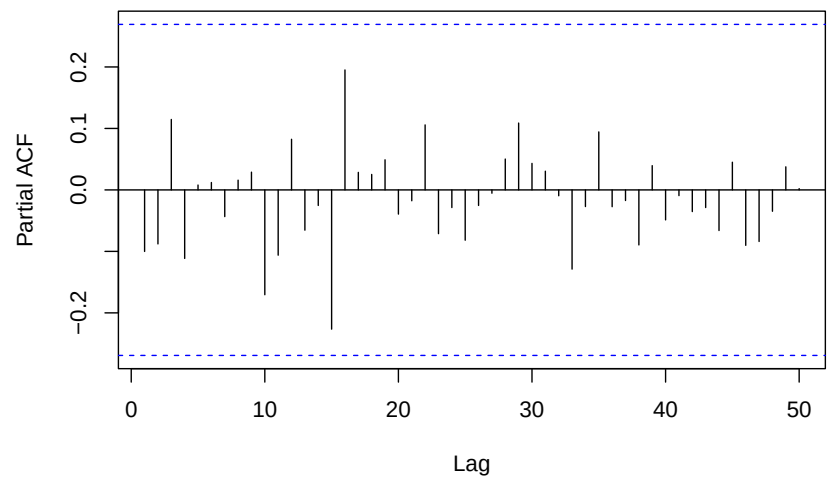


```
## Series: TLB_ts
## ARIMA(14,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.0260  0.0380 -0.0184 -0.0555  0.0794  0.1459  0.1111  0.1666  0.0608
## s.e.  0.1344  0.1218  0.1091  0.1027  0.1050  0.1015  0.1087  0.1051  0.1087
##      ar10     ar11     ar12     ar13     ar14
##      -0.0696 -0.2211  0.2823 -0.3715 -0.2759
## s.e.  0.1085  0.1078  0.1195  0.1245  0.1466
##
## sigma^2 = 5629775: log likelihood = -472.66
## AIC=975.33  AICc=988.66  BIC=1004.6
```

ACF of Residuals for ARIMA(14,1,0) for TLB



PACF of Residuals for ARIMA(14,1,0) for TLB

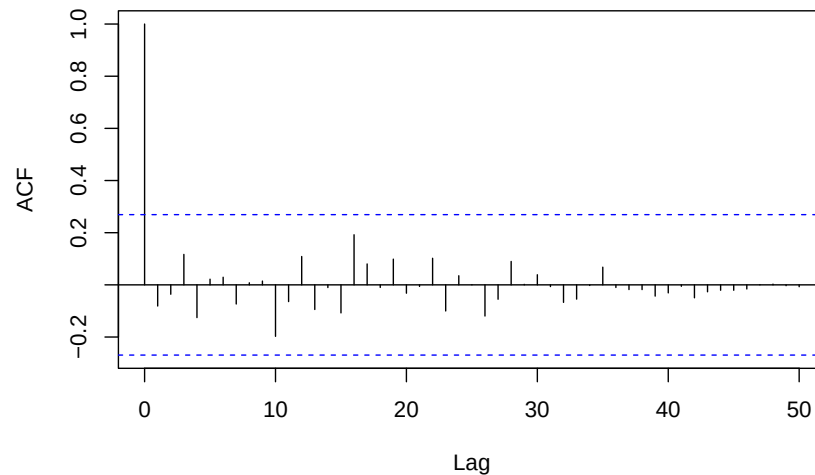


```

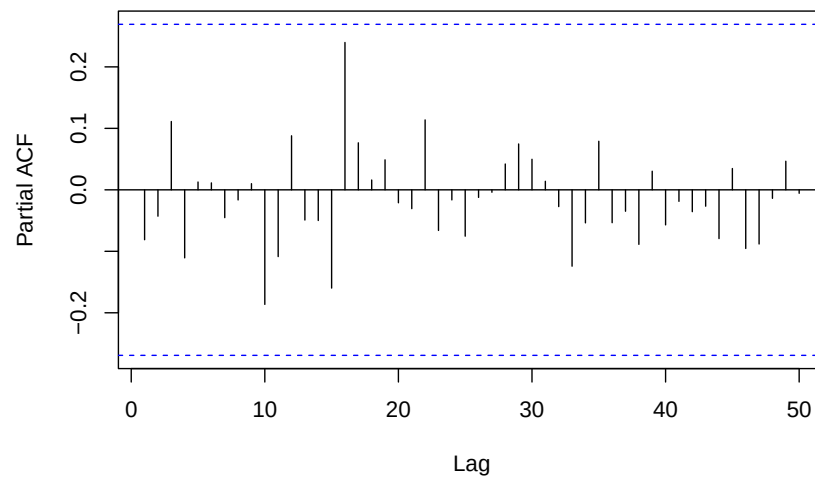
## Series: TLB_ts
## ARIMA(13,1,1)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##      0.5571 -0.0703  0.0264 -0.0762  0.0940  0.1004  0.0542  0.1077 -0.0216
## s.e.  0.1677  0.1156  0.1116  0.1120  0.1104  0.1040  0.1090  0.1095  0.1125
##      ar10     ar11     ar12     ar13     ma1
##      -0.0853 -0.1964  0.4091 -0.5605 -0.5344
## s.e.   0.1130   0.1112  0.1223  0.1126  0.1886
##
## sigma^2 = 5503186: log likelihood = -472.28
## AIC=974.57  AICc=987.9  BIC=1003.83

```

ACF of Residuals for ARIMA(13,1,1) for TLB



PACF of Residuals for ARIMA(13,1,1) for TLB

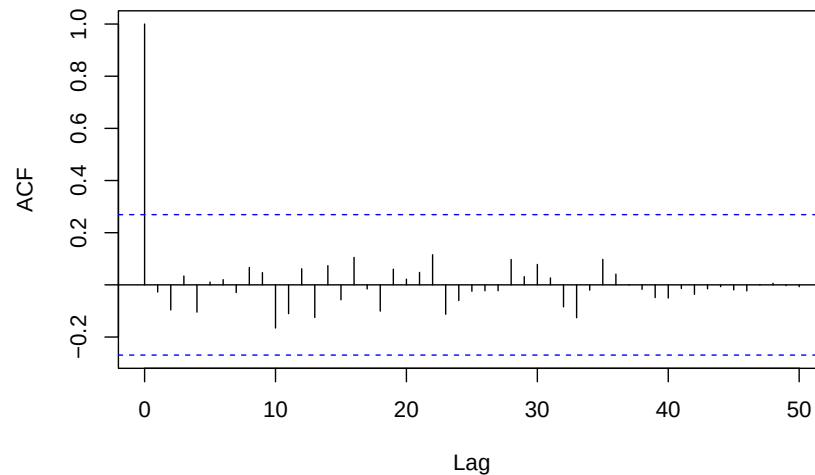


```

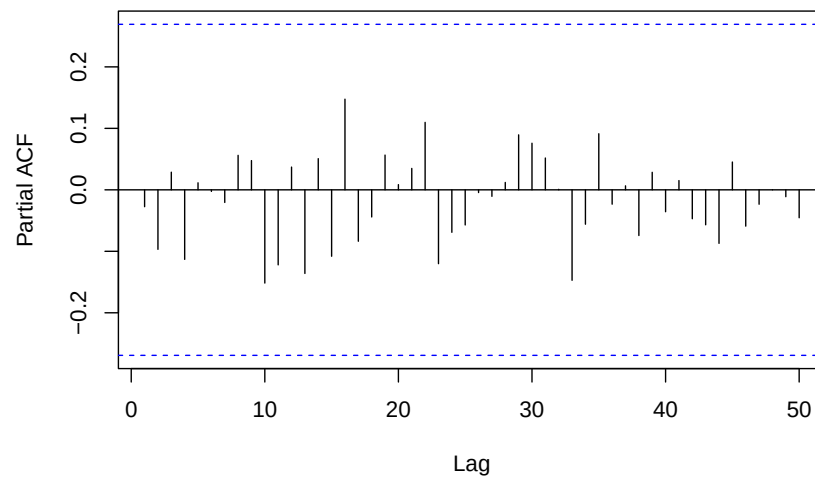
## Series: TLB_ts
## ARIMA(13,1,2)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##      0.6648 -0.4027 -0.0077 -0.0835  0.1408  0.0861  0.0430  0.1468
## s.e.  0.1376  0.1968  0.1260  0.1252  0.1252  0.1203  0.1199  0.1246
##      ar9      ar10     ar11     ar12     ar13      ma1      ma2
##      -0.0053 -0.0478 -0.1305  0.4076 -0.6385 -0.7326  0.5855
## s.e.   0.1262  0.1271  0.1253  0.1159  0.1041  0.1611  0.3171
##
## sigma^2 = 5122169: log likelihood = -470.51
## AIC=973.01  AICc=988.56  BIC=1004.23

```

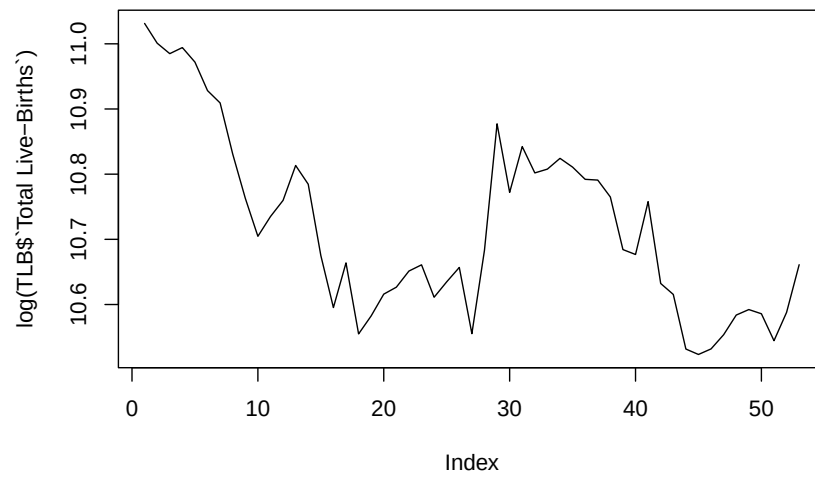
ACF of Residuals for ARIMA(13,1,2) for TLB



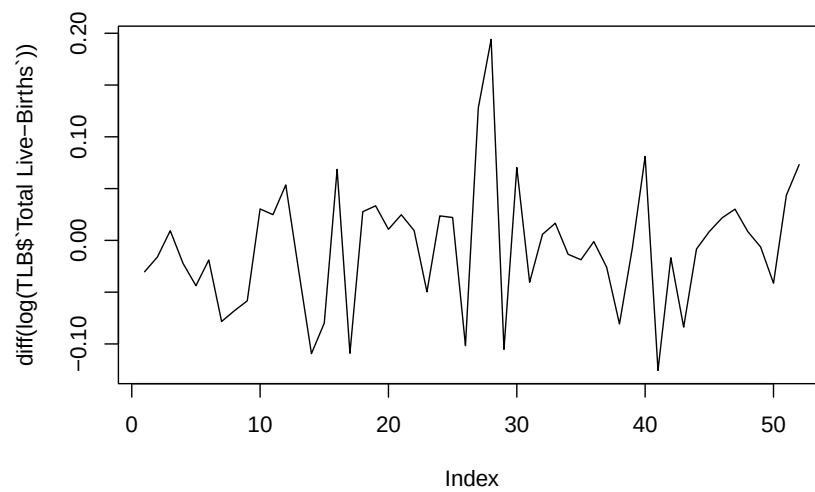
PACF of Residuals for ARIMA(13,1,2) for TLB



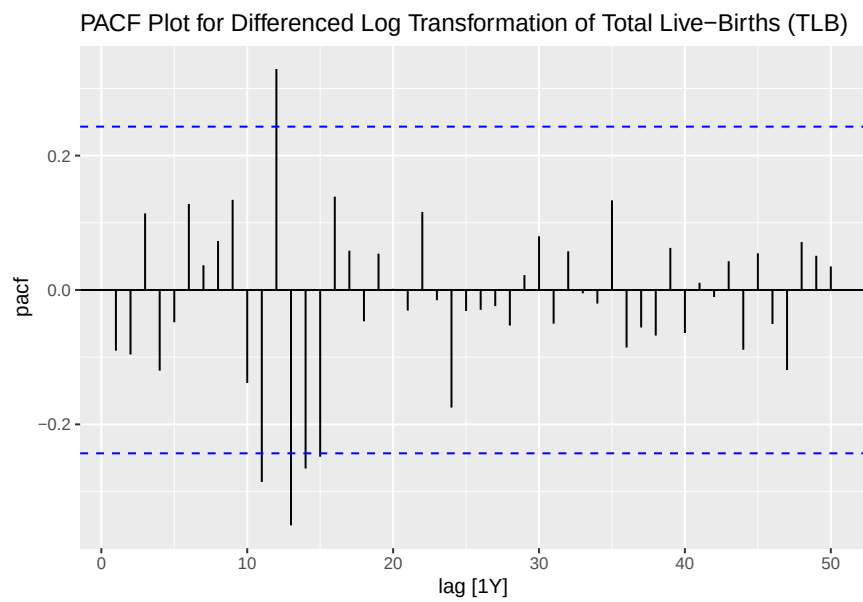
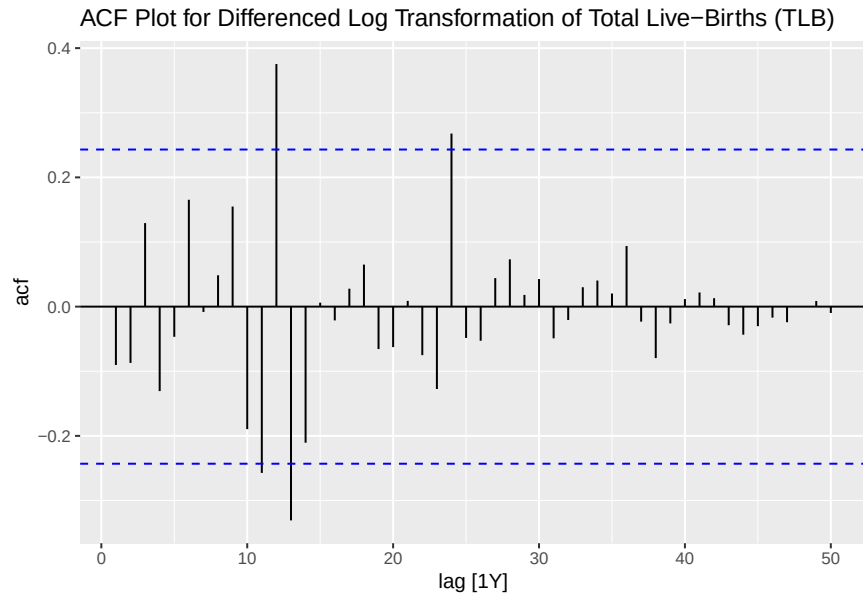
Plot of Log Transformation for Total Live-Births (TLB) until 2012



Plot of Differenced Log Transformation for Total Live-Births (TLB) until

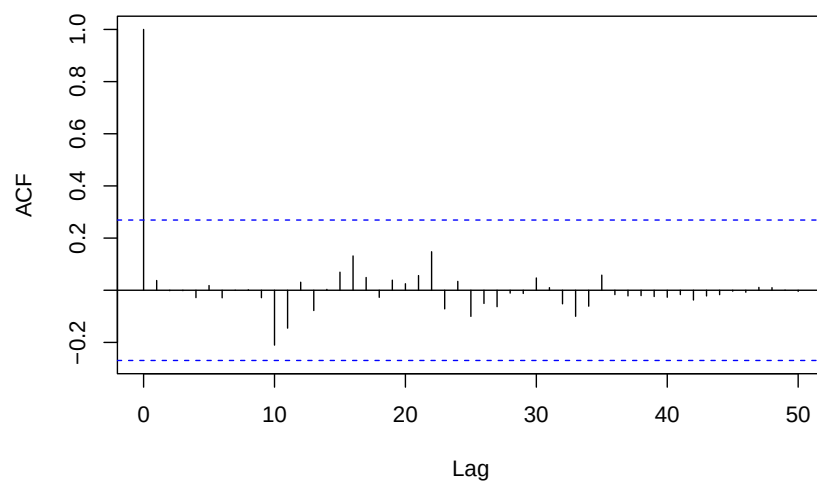


```
##      kpss_stat kpss_pvalue
##      0.1765042  0.1000000
```

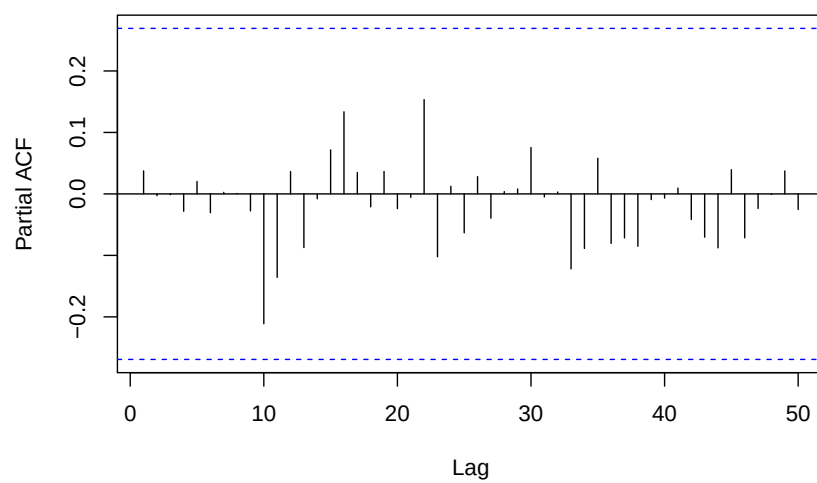


```
## Series: log(TLB_ts)
## ARIMA(15,1,0)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8      ar9
##    -0.1455 -0.1154  0.0919 -0.1862  0.0220  0.1559  0.1565  0.1742  0.0826
## s.e.  0.1315  0.1197  0.1028  0.0946  0.0935  0.0868  0.0972  0.0918  0.0931
##      ar10     ar11     ar12     ar13     ar14     ar15
##    -0.0706 -0.2325  0.2774 -0.3642 -0.3436 -0.4237
## s.e.  0.0939  0.0892  0.0988  0.1089  0.1264  0.1359
##
## sigma^2 = 0.00225: log likelihood = 88.78
## AIC=-145.56  AICc=-130.02  BIC=-114.34
```

ACF of Residuals for ARIMA(15,1,0) for TLB with log transformation



PACF of Residuals for ARIMA(15,1,0) for TLB with log transformation

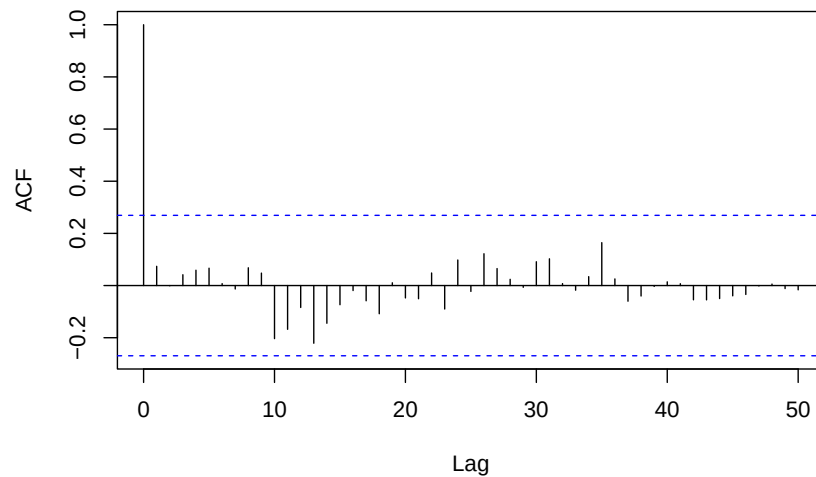


```

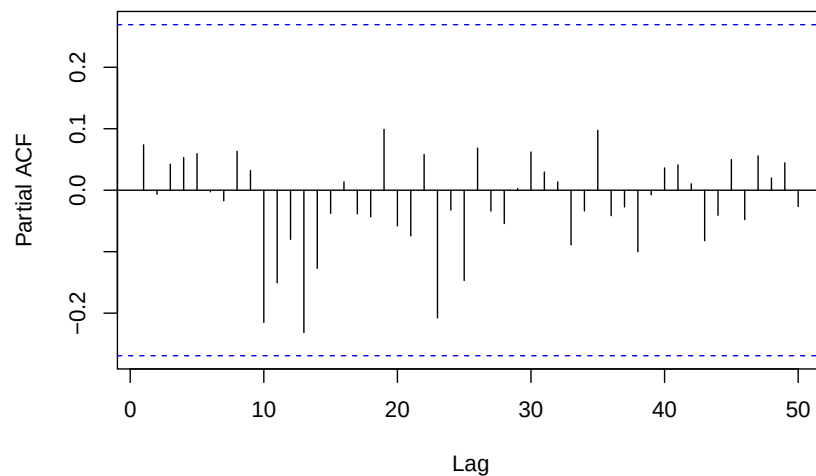
## Series: log(TLB_ts)
## ARIMA(12,1,3)
##
## Coefficients:
##      ar1      ar2      ar3      ar4      ar5      ar6      ar7      ar8
##    -0.3816 -0.3915 -0.4801 -0.1650 -0.0459  0.0622  0.0321  0.1705
## s.e.   0.1314   0.1535   0.1615   0.1495   0.1471   0.1454   0.1494   0.1589
##      ar9      ar10     ar11     ar12     ma1      ma2      ma3
##      0.1587  0.0805  -0.0368  0.4897  0.5443  0.4755  0.9169
## s.e.   0.1683  0.1646   0.1479  0.1451  0.1011  0.2255  0.2718
##
## sigma^2 = 0.002424: log likelihood = 85.85
## AIC=-139.69  AICc=-124.15  BIC=-108.47

```

ACF of Residuals for ARIMA(12,1,3) for TLB with log transformation



PACF of Residuals for ARIMA(12,1,3) for TLB with log transformation



Autoregressive Model Integrated (ARIMA) model

$$\phi(B)(1-B)^d y_t = c + \theta(B)\varepsilon_t$$

Table 5. Full table for TFR - no log transformation

Model	AIC	AICc	MSE	MAE	
ARIMA(16,1,0)	-63.99	-45.99	0.04753075	0.1690442	Lowest AIC
ARIMA(16,1,1)	-62.39	-42.66	0.04783471	0.1700046	
ARIMA(15,1,1)	-60.26	-42.26	0.08431368	0.223524	
ARIMA(14,1,3)	-60.99	-40.26	0.01899351	0.09964934	
ARIMA(12,2,3)	-62.57	-46.57	0.01218546	0.08274547	Lowest AICc
ARIMA(11,2,3)	-52.91	-39.19	0.05260825	0.1823268	
ARIMA(10,2,3)	-54.69	-43.02	0.05680021	0.1898403	

Table 6. Full table for TFR - log transformed

Model	AIC	AICc	MSE	MAE	
ARIMA(13,1,3)	-136	-118	0.02602064	0.121592	Lowest AIC Log Transform
ARIMA(12,1,4)	-131.83	-113.83	0.02101819	0.1132223	
ARIMA(13,1,5)	-133.96	-110.21	0.05826379	0.1840232	
ARIMA(12,1,5)	-130.02	-109.3	0.02172126	0.1147922	
ARIMA(15,2,0)	-133.67	-117.67	0.01869052	0.1056717	
ARIMA(15,2,2)	-135.54	-114.17	0.02235162	0.1187815	
ARIMA(14,2,2)	-130.49	-111.94	0.01087075	0.08352761	
ARIMA(13,2,2)	-132.39	-116.39	0.007426324	0.06826933	Lowest MSE & MAE
ARIMA(12,2,2)	-127.34	-113.62	0.0498791	0.1720094	

Table 7. Full table for TLB - no log transformation

Model	AIC	AICc	MSE	MAE	
ARIMA(14,1,0)	975.33	988.66	67367429	6853.008	
ARIMA(14,1,1)	975.62	991.17	84179161	7769.997	
ARIMA(13,1,1)	974.57	987.9	82856991	7698.072	lowest AICc (no log trans- formation)
ARIMA(13,1,2)	973.01	988.56	54382525	6014.802	
					lowest AIC (no log trans- formation), MSE & MAE

Table 8. Full table for TLB - log transformed

Model	AIC	AICc	MSE	MAE	
ARIMA(15,1,0)	-145.56	-130.02	2239721240	47245.34	lowest AICc (log transformed)
ARIMA(15,1,1)	-145.43	-127.43	2173041601	46542.86	
ARIMA(14,1,1)	-140.56	-125.02	2211919788	46958.53	
ARIMA(13,1,1)	-141.45	-128.11	2200303964	46836.59	
ARIMA(12,1,3)	-139.69	-124.15	1752591080	41851.92	

<https://github.com/twalepear/TSA.git>